1 a) $y' = \frac{1}{2x}$

$\frac{y'}{y} = \frac{1}{2x}$

$\frac{2y'}{y} = \frac{1}{x}$

$\int \frac{2\,dy}{y} = \int \frac{dx}{x}$

$2\ln(y) = \ln(x) + C$

$e^{2\ln(y)} = e^{\ln(x)+C}$

$y^2 = x \cdot e^C$

$y = \sqrt{x} + D$ pro $x = \mathbb{R}$

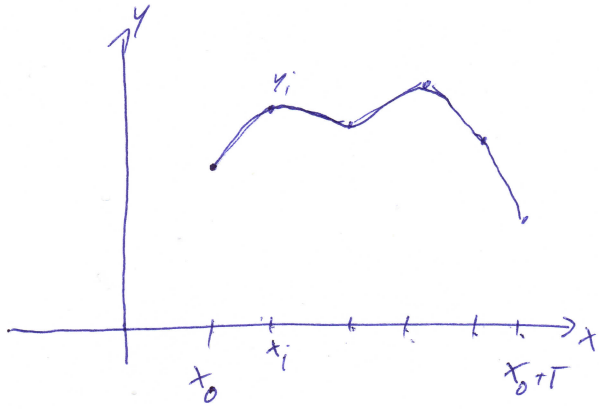
$y(1) = 1$

$1 = \sqrt{1} + D$

$D = 0$

$y = \sqrt{x}$

b) a)



$x_i = x_0 + i \cdot \frac{T}{n}$

$y_i \approx y(x_i) = y\left(x_0 + i \cdot \frac{T}{n}\right)$

b) $x_i = 1 + i \cdot \frac{5}{5} = 1 + i$; $i \in \langle 0; 5 \rangle$

$y_i \approx y(1+i)$; $i \in \langle 0; 5 \rangle$

$y_i = y_{i-1} + k \cdot \frac{T}{n}$

j1)

$x_1 = 1 + 1 = 2$

$k = \frac{y}{2x}$ $k_0 = \frac{1}{2}$

$y_1 = y_0 + k \cdot h$

$y_1 = 1 + \frac{1}{2} \cdot 1$

$y_1 = \frac{3}{2}$

$x_2 = 1+2 = 3$ $\qquad k_1 = \frac{\frac{3}{2}}{4} = \frac{3}{8}$

$y_2 = \frac{3}{2} + \frac{3}{8} = \frac{15}{8}$

$x_3 = 1+3 = 4$ $\qquad k_2 = \frac{\frac{15}{8}}{6} = \frac{5}{16}$

$y_3 = \frac{15}{8} + \frac{5}{16} = \frac{35}{16}$

2

a)

$E_{0,5} = c \cdot 0,5^2$ $\qquad \frac{0,5}{0,1} = 5$

$0,01 = c \cdot 0,25$

$0,04 = c$ $\qquad E_{0,1} = \frac{1}{25} E_{0,5} = \frac{0,01}{25} = \underline{\underline{0,0004}}$

b)

$0,0025 = c \cdot h^2$

$0,0025 = 0,04 \cdot h^2$

$\frac{1}{16} = h^2$

$\underline{\underline{\frac{1}{4} = h}}$